

ConSight — Post-Phase 4 Add-On Implementation Plan

SIH26122 | Internal Hackathon MVP | Oil India Limited

Purpose: Close the most important gaps after Phase 4 and prepare a strong, reliable SIH internal-hackathon prototype.

1. Objective

Phase 1–4 is treated as the existing core MVP. The add-on work below is intentionally limited to features that directly strengthen SIH26122 alignment and create a convincing judge demonstration. The source analysis identifies the complete value pipeline as field capture → AI extraction → L5/L6 matching → confidence → planner review → audit trail → schedule update → delay impact → institutional memory.

2. Required Add-On Phases

Phase	Feature	Priority	Judge Impact	What to Implement
Phase 5	Primavera P6 / MS Project Round-Trip	MUST HAVE	Critical	Import P6/MS Project schedule data; map activities into ConSight; update actual start/end dates; export updated schedule data.
Phase 6	WhatsApp Field Reporting	MUST HAVE	Very High	Allow supervisors to send field updates through WhatsApp and route them through the existing extraction/matching pipeline.
Phase 7	Multi-Language / Hinglish Extraction	REQUIRED	High	Handle realistic Indian code-mixed field messages such as Hindi-English and map them to English L5/L6 activities.
Phase 8	Delay Ripple / Critical Path Impact	REQUIRED	High	When a delay is logged, show affected downstream activities and critical-path impact using schedule dependencies.
Phase 9	Productivity & Institutional Memory Analytics	REQUIRED	High	Turn accumulated audit/progress data into useful trends:

				actual vs planned, discipline productivity, recurring delay patterns and variance.
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3. Phase 5 — P6 / MS Project Round-Trip

Priority: #1 — this is the most important missing SIH requirement.

- Input: Primavera P6 export (.xer) and, if feasible, MS Project export.
- Extract: activity ID, activity name/description, planned dates, actual dates, discipline, predecessor/successor relationships and other fields required by the matching engine.
- Map the imported schedule into the existing ConSight L5/L6 activity model.
- After a field event is approved, update the corresponding actual start/end date.
- Provide an export/download path that demonstrates the updated schedule data going back to the scheduling workflow.
- Demo target: Import schedule → field event → match → approve → actual date changes → export.

4. Phase 6 — WhatsApp Field Reporting

Priority: #2 — strongest 'minimal friction' demonstration.

- Connect WhatsApp Business Cloud API/webhook to the existing FastAPI backend.
- Treat WhatsApp as another input channel; reuse the existing extraction, matching and confidence services.
- Accept simple text progress messages from supervisors.
- Show the received message, extracted event, matched L5/L6 activity, confidence and review status on the dashboard.
- Demo target: Supervisor sends one WhatsApp message → ConSight processes it → planner sees the matched activity.

5. Phase 7 — Multi-Language / Hinglish Extraction

Priority: #3 — low implementation effort with strong India-specific demo value.

- Update extraction prompts to understand code-mixed field language.
- Support realistic Hindi-English examples first; extend to other required languages only if reliable.
- Preserve activity IDs/codes and construction terminology while extracting structured fields.
- Create and test a benchmark set of roughly 20–30 realistic code-mixed messages.
- Demo target: Hinglish message → correct structured event → correct English L5/L6 match.

6. Phase 8 — Delay Ripple / Critical Path Impact

Priority: #4 — visual proof that field events affect the schedule.

- Use predecessor/successor relationships imported from the schedule.
- When a delay event is approved, traverse affected successor activities.

- Calculate and display the immediate downstream impact based on available schedule relationships.
- Clearly show whether the critical path is affected.
- Avoid building a complex predictive ML model for the internal MVP; use deterministic schedule dependency analysis.
- Demo target: Activity delayed by 2 days → affected downstream activities shown → critical-path warning displayed.

7. Phase 9 — Productivity & Institutional Memory Analytics

Priority: #5 — completes the long-term value story.

- Use the existing audit/progress database as the source.
- Show planned vs actual duration and variance.
- Show productivity/performance by discipline and activity type.
- Show recurring delay causes/patterns where data exists.
- Show variance trends over time.
- Keep the implementation structured and queryable; advanced RAG is not required for the SIH MVP.
- Demo target: Historical execution data → useful patterns/insights → evidence of growing institutional memory.

8. Internal Hackathon Definition of Done

- ☐ Phase 1–4 three-case demo works reliably: clean match, ambiguous match with planner correction, and unmatched activity/create-new workflow.
- ☐ P6 schedule can be imported and mapped to ConSight L5/L6 activities.
- ☐ Approved actual dates can be demonstrated as flowing back into the schedule/export.
- ☐ At least one WhatsApp field-report flow works end-to-end.
- ☐ At least 20–30 Hinglish/code-mixed test messages have been evaluated.
- ☐ At least one delay-ripple scenario works using real schedule dependencies.
- ☐ Insights page shows institutional-memory/productivity analytics from stored execution data.
- ☐ Confidence score and audit trail remain visible for every important AI decision.
- ☐ A fallback/demo dataset or mock mode exists in case an external API fails during judging.

9. Do NOT Add Before Internal SIH

- BIM / Digital Twin
- Weather / external-context integrations
- Native mobile application
- Multi-project enterprise architecture
- Advanced RAG system
- Complex ML-based delay prediction
- Voice processing unless the core pipeline is already stable

- OCR of scanned diaries as a core live-demo dependency

10. Recommended Build Order

1. Test and stabilize Phase 1–4.
2. Build Phase 5 — P6/MS Project round-trip.
3. Build Phase 6 — WhatsApp integration.
4. Add Phase 7 — Hinglish/multi-language extraction and benchmark testing.
5. Build Phase 8 — delay ripple / critical-path impact.
6. Complete Phase 9 — productivity and institutional-memory analytics.
7. Run full end-to-end demo testing and prepare a fallback demo dataset.

11. Final MVP Story for Judges

ConSight should demonstrate one connected workflow: a site supervisor reports progress through a low-friction channel; AI extracts the activity event; the system fuzzy-matches it to the correct L5/L6 schedule node with a confidence score; the planner can approve or correct it; the actual date is updated in the schedule workflow; downstream delay impact is shown; and the event becomes part of a growing, queryable institutional-memory dataset.

Source basis: ConSight — Feature Analysis & Recommendation Rationale.pdf, provided for SIH26122 planning. This plan preserves the source document's five SIH requirements and its recommended post-Phase-4 priorities.